TECHNICAL SCOPE / TOR BRIEF

Lesotho Child Helpline 116 / OpenCHS - Amendment 3

Prepared for Inception Meeting - 24 September 2026

Contract	UNICEF Lesotho Contract No. 43427866 - Amendment 3
Contractor	BITZ IT Consulting Ltd
Contract validity	10 March 2025 to 7 July 2027
Primary focus	OpenCHS integration, configuration, testing, training and operational handover

Source basis: UNICEF institutional contract 43427866, Amendment 3, including the Statement of Work/TOR and revised service items.

1. Overall Purpose

The assignment strengthens the Lesotho Child Helpline 116 / OpenCHS environment through requirements definition, system configuration, outbound API integration, testing, user acceptance, capacity building and go-live support. The broader TOR objective is to ensure optimal operation of the Child Helpline, build staff capacity and provide system support and maintenance to enhance child protection.

2. Revised Technical Scope - Four Workstreams

Workstream	Scope	Primary Output
1. Inception, Requirements & Environment Setup	Confirm business and technical requirements, architecture, environments, access, integration workflow and implementation plan.	Inception / requirements / environment setup package
2. ABR ID + Validation + Referral UI	Configure the ABR ID field, validation rules and referral checkbox/interface in OpenCHS.	Configured fields, rules and referral user interface
3. Outbound REST API Client	Develop outbound API integration with JWT authentication, retries, offline queuing, idempotency and audit logging.	Working and documented outbound integration
4. Testing, UAT, Training & Go-Live	Execute technical testing and UAT, train in-country operators/IT staff, deploy and hand over.	UAT evidence, training, go-live and handover

3. Workstream 1 - Inception, Requirements & Environment Setup

The inception phase should create a shared technical baseline before development begins. The team should document and confirm:

- Current OpenCHS architecture and the Lesotho deployment environment.
- The external child-protection / case-management platform to be integrated with OpenCHS.
- Which system initiates the referral and the exact end-to-end referral workflow.

- Data to be transferred, including mandatory/optional fields and field mapping.
- ABR ID ownership, creation point, format, uniqueness and validation rules.
- User roles, permissions and who is authorized to initiate referrals.
- Development, test/UAT and production environments and access arrangements.
- API endpoints, connectivity, authentication and security requirements.
- Logging, audit, retry, offline and error-handling expectations.
- UAT acceptance, sign-off, go-live and support arrangements.

4. Workstream 2 - ABR ID Field, Validation Rules & Referral Checkbox UI

Technical scope

- Add or configure the ABR ID in the required OpenCHS form/workflow.
- Confirm whether the ABR ID is generated internally, entered manually or returned/provided by another system.
- Implement field type, length, format, required-field and uniqueness rules as agreed.
- Provide clear validation messages where entered data is missing or invalid.
- Provide a referral checkbox/control and define who can initiate a referral.
- Ensure referral conditions are satisfied before submission and prevent accidental or duplicate referral where required.
- Display referral status clearly to operators.

Questions to settle at inception

1. What exactly does ABR stand for in the receiving system?

2. Who creates the ABR ID and at what stage?

3. Is it nationally unique or system-specific?

4. What is the required format?

5. What happens where no ABR ID is available?

6. Can a referral be cancelled or corrected?

7. Can the same case be referred more than once?

8. Who is authorized to select the referral checkbox?

5. Workstream 3 - Outbound REST API Integration

The revised scope specifically calls for an outbound REST API client with JWT authentication, retry logic, offline queueing, idempotency and audit logging.

A. REST API client

- API base URL and available endpoints.
- HTTP methods, headers, request payload and response payload.
- Error codes and error-response structure.
- API version, sandbox/UAT endpoint and production endpoint.

B. JWT authentication

- Token issuer and authentication endpoint.
- Required credentials, certificates or secrets.
- Token lifetime, expiry and refresh process.

- Secure storage of credentials/secrets.
- Separate test and production credentials where applicable.

C. Data mapping

Prepare a formal mapping table: OpenCHS field -> API field -> validation/transformation rule. The mapping must be confirmed jointly; it should not be assumed by developers.

D. Retry mechanism and offline queue

- Define which failures trigger retries and the retry/back-off strategy.
- Define maximum retry attempts and escalation after exhaustion.
- Retain failed/unavailable transmissions so referrals are not lost.
- Provide queue monitoring and, where appropriate, controlled manual retry.

E. Idempotency

The integration must prevent duplicate cases/referrals when retries occur. A unique idempotency / transaction key must therefore be agreed with the receiving-system team.

F. Audit logging

- Case/reference ID and ABR ID where available.
- Transaction ID, date/time and initiating user/system.
- Endpoint/action invoked and transmission status.
- Success/failure response and retry history.
- Logging should avoid unnecessary exposure of sensitive child-protection data.

6. Workstream 4 - Testing, UAT, Training & Go-Live

Technical and regression testing

- Field and validation testing.
- Authentication and API connectivity testing.
- Successful referral and failed referral scenarios.
- Timeout, retry and offline queue behaviour.
- Duplicate/idempotency testing.
- Audit logging and access-control testing.
- Regression testing to ensure existing OpenCHS functionality is not negatively affected.

Suggested UAT scenarios

1. Create/update a case.

2. Enter or obtain ABR ID.

3. Select referral and validate mandatory information.

4. Transmit and confirm successful receipt by the target platform.

5. Test rejected/invalid referral.

6. Test API unavailability and queued referral.

7. Retry transmission and confirm no duplicate is created.

8. Verify referral status and audit trail.

Training and handover

Operator/social-worker training should cover the new ABR ID and referral workflow, validation messages, referral status, failure handling and escalation. Technical/IT training should cover architecture, API configuration, authentication, monitoring, logging, queue management, retries, troubleshooting, environment configuration and support escalation.

7. Existing Support Obligations That Continue

The integration sits within a broader support-and-maintenance TOR. Existing responsibilities include:

- Minor system updates that do not require major module overhaul.
- Bug fixing, onsite support and knowledge transfer.
- Creating/availing API endpoints for partner integrations.
- Server software and dependency maintenance/updates.
- Quality assurance after changes and dependency updates.
- Training of IT personnel.
- Maintaining system stability and functionality.
- Monitoring and recommendations on hardware resource utilization.
- Support with third-party service providers where required.
- Consultancy advice on system improvements and gathering user feedback for future upgrades.

8. Inputs Required From UNICEF / MGYSD / Receiving-System Team

Required input / decision	Likely owner
Business referral workflow	MGYSD
ABR definition and business rules	MGYSD / receiving platform
API specification/documentation	Receiving-system technical team
Sandbox/test API URL	Receiving-system technical team
JWT/authentication documentation	Receiving-system technical team
Sample request/response payloads	Receiving-system technical team
Data dictionary / code lists	Both system teams
Field mapping	Joint
Mandatory fields and validation rules	MGYSD
UAT participants and sign-off authority	MGYSD / UNICEF
Production endpoint and credentials	Receiving-system technical team
Network/firewall/VPN requirements	IT teams
Data protection/security constraints	UNICEF / MGYSD
Go-live date	Joint

9. Architecture to Confirm During Inception

Child Helpline Operator

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OpenCHS Case Management

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ABR ID + Validation + Referral Trigger

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Outbound Integration Service

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JWT Authentication

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REST API

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Child Protection Case Management System

Supporting controls: Retry Engine -> Offline Queue -> Idempotency Control -> Audit Log

A key inception decision is whether the integration is strictly outbound, as the revised service description indicates, or whether any status/response information must also be written back into OpenCHS. Any bi-directional requirement should be explicitly confirmed and documented.

10. Decisions / Actions to Close Today

1. Integration systems and owners confirmed.

2. End-to-end referral workflow agreed.

3. ABR ID definition, ownership and rules agreed.

4. Data fields for transmission agreed or field-mapping exercise assigned.

5. API documentation shared.

6. Test environment and credential process agreed.

7. Authentication and security requirements agreed.

8. Error, retry, offline and idempotency behaviour agreed.

9. UAT participants and acceptance criteria agreed.

10. Development -> testing -> UAT -> training -> go-live timetable agreed.

11. Contractual Deliverables - Team Focus

D1 - Inception, Requirements & Environment Setup

D2 - ABR ID Field + Validation Rules & Referral Checkbox UI

D3 - Outbound REST API Client: JWT + Retry + Offline Queue + Idempotency + Audit Logging

D4 - Testing + UAT + In-Country Operator Training + Go-Live / Handover

Meeting emphasis: The inception meeting should primarily close requirements, interfaces, responsibilities, dependencies and acceptance criteria so the technical team can proceed without assumptions. The immediate outputs should be the Inception Report, Requirements Specification, Integration Design / Field-Mapping document, and a clear action register for unresolved items.